VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE ADVANCED - 5 | PAPER - 2

JEE 2022

Date	Maximam Marks	Duration
2nd January, 2022	180	3 Hours

General Instructions

1. The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).

2. **Section 1** contains **2 types** of questions **[Type A and Type B].**

   Type A contains **SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B),(C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

   Type B contains **THREE (03) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

3. **Section 2** contains **6 Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

4. For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) : ______________________

Roll Number : ______________________

OMR Bar Code Number : ______________________

Candidate's Signature : ____________ Invigilator's Signature ____________

MARKING SCHEME

SECTION-1 | Type A

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen;
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If unanswered;
  Negative Marks : −2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
  choosing ONLY (A), (B) and (D) will get +4 marks;
  choosing ONLY (A) and (B) will get +2 marks;
  choosing ONLY (A) and (D) will get +2marks;
  choosing ONLY (B) and (D) will get +2 marks;
  choosing ONLY (A) will get +1 mark;
  choosing ONLY (B) will get +1 mark;
  choosing ONLY (D) will get +1 mark;
  choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
  choosing any other option(s) will get −2 marks.

SECTION-1 | Type B

- This section contains **THREE (03) paragraphs**. Based on each paragraph, there are **TWO (02)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme;

  Full Marks : +3 If ONLY the correct option is chosen;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : −1 In all other cases.

SECTION - 2

- This section contains **6 Integer Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

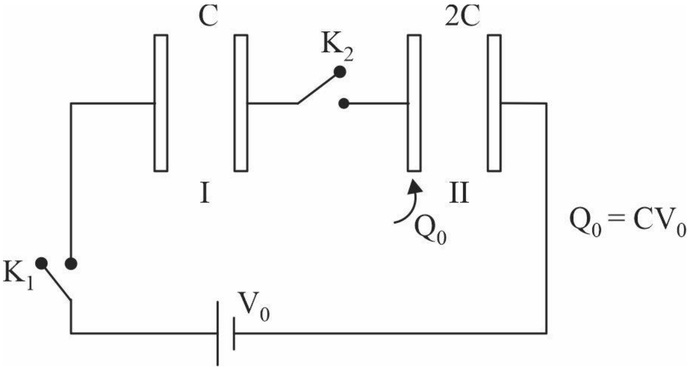
  Full Marks: +3 If ONLY the correct Integer value is entered. There is **NO negative marking.**
  Zero Marks: 0 In all other cases.

SUBJECT I : PHYSICS 60 MARKS

SECTION-1 | Type A

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

1. Two parallel plate capacitors with capacitance C and 2C are connected in series. Capacitor with capacitance C is initially uncharged and left plate of capacitor with capacitance 2C has charge Q_0. Both the keys are closed simultaneously. Choose the correct statement(s).



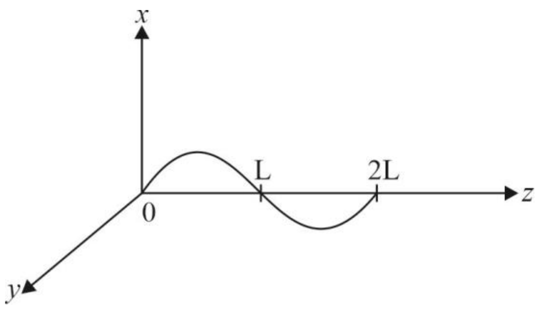
(A) Final charge on capacitor I is $\dfrac{2CV_0}{3}$

(B) Charge supplied by battery after closing the switches is $\dfrac{CV_0}{2}$

(C) Ratio of final potential difference across the two capacitors is 1

(D) Change in electrostatic energy stored in capacitor II after closing the key is $\dfrac{3CV_0^2}{16}$

2. A wire carrying current I is kept in *x*–*z* plane with its shape given by $x = A\sin\left(\dfrac{\pi z}{L}\right)$; $0 \le z \le 2L.$ A uniform magnetic field of magnitude B exists in the region. Force acting on the wire is:



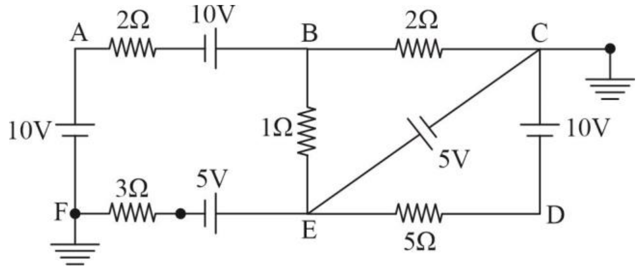
(A) Zero if B is along *y*-direction

(B) Zero if B is along *z*-axis

(C) 2IBL if B is along *x*-axis

(D) 2ABL if B is along *z*-axis

3. In the circuit shown below, choose the correct statements.



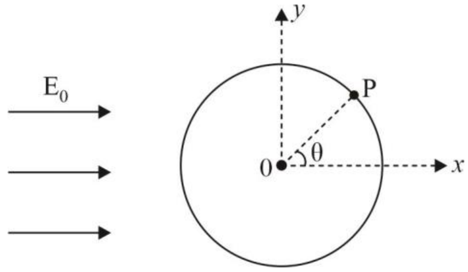
(A) Current through 3Ω resistance is zero

(B) Current through 1Ω resistance is 2.5A

(C) Current in branch CE is 0.5A

(D) Current though branch AB is zero

4. A neutral conducting sphere of radius R with centre at origin is placed in a uniform electric field $E_0\hat{i}$ due to which charge is induced on the surface of the conductor.



At a point P (in x-y plane) on the conductor local surface charge density is found to be σ. OP makes an angle θ with *x*-axis. Choose the correct statement(s):

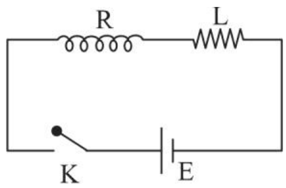
(A) Just outside the point P electric field is greater than $\frac{\sigma}{\epsilon_0}$

(B) Electric field due to induced charges on the sphere just outside P is normal to the surface

(C) If a small area *dA* is taken at P force on charge present in this small area will be $\frac{\sigma^2 dA}{2\epsilon_0}$

(D) For $\theta = 90°$ electric field just outside P, due to induced charges on sphere is $\left(E_0^2 + \left(\frac{\sigma}{\epsilon_0}\right)^2\right)^{1/2}$

5. An LR circuit with a cell of emf E is switched on at t = 0. Consider t_0 is the time when current through the inductor is half of its steady state value and t_0' be the time when energy stored in the inductor is half of its steady state value. Choose the correct statement(s) :



(A) During $t = 0$ to $t = t_0$, charge flown through the battery is $\frac{EL}{2R^2}(\ln(4) - 1)$

(B) At $t = t_0$; rate at which energy is transferred to inductor is maximum

(C) $t_0' = 2t_0$

(D) $t_0' - t_0 = \frac{L}{R}\ln\left(\frac{1}{2-\sqrt{2}}\right)$

6. A capacitor with capacitance $C = 1\mu F$ and a wire of resistance $R = 10\Omega$ and inductance L = 1.0 mH are connected in parallel to a source of sinusoidal voltage V = 31.6 volts. Choose the correct option(s). [Take $\sqrt{10} = 3.16, \sqrt{5} = 2.24$]

(A) Frequency ω at which resonance sets in is $3.16 \times 10^4 \frac{\text{rad}}{s}$

(B) Current through the inductor is 1A at resonance

(C) Current through the capacitor is 2A at resonance

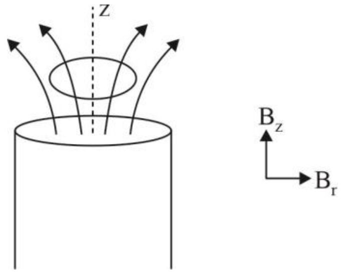
(D) Current through the source is 3.16A at resonance

SECTION-1 | Type B

This section consists of THREE (03) paragraphs. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question 7 - 8

A thin superconducting ring is held above a vertical, cylindrical magnetic rod. The axis of symmetry of the ring is the same as that of the rod. The cylindrically symmetrical magnetic field around the ring can be described approximately in terms of vertical and radial components of magnetic field vector $B_z = B_0(1-\alpha z)$ & $B_r = B_0\beta r,$ where B_0, α, β are positive constants, z and r are vertical and radial position coordinates respectively. Initially ring has no current flowing in it. When released it starts to move downwards with its axis still vertical and it performs harmonic oscillations for small displacement. Initially $z = 0$ for ring. Mass of the ring is m, radius is r and its inductance is L.



7. Current in the ring at a displacement z from initial position is:

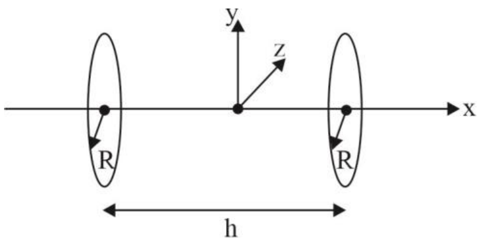
(A) $\dfrac{B_0 r_0^2 \pi z}{2L}$ **(B)** $\dfrac{2B_0\alpha\pi r_0^2 z}{L}$ **(C)** $\dfrac{B_0\alpha\pi r_0^2 z}{L}$ **(D)** $\dfrac{B_0\alpha r_0^2 z}{2L}$

8. Ring makes harmonic oscillations with amplitude:

(A) $\dfrac{mgL}{2B_0\pi^2 r_0^2\alpha\beta}$ **(B)** $\dfrac{2mgL}{B_0^2\pi r_0^4\alpha\beta}$ **(C)** $\dfrac{mg\beta}{2B_0\pi r_0^2\alpha L}$ **(D)** $\dfrac{mgL}{2B_0^2\pi^2 r_0^4\alpha\beta}$

Paragraph for Question 9 - 10

A helmholtz coil is a device for producing a region of nearly uniform magnetic field. Besides creating magnetic fields, helmholtz coils are used in scientific apparatus to cancel external magnetic fields, such as earth's magnetic field. A helmholtz pair consists of two identical circular magnetic coils that are placed symmetrically along a common axis, one on each side of experimental area and separated by a distance equal to radius R of the coil. Each coil carries equal current in the same direction. Setting the distance between the coils $h = R,$ is what defines a Helmholtz pair, it minimizes the non-uniformity of the field at the centre of the coils such that the first non-zero derivative is $\dfrac{d^4B}{dx^4}$.



The calculation of exact magnetic field at any point in space is very complex. But field at any point on the axis can be calculated easily.

9. If number of turn in each coil is n and current is I then magnetic field at the midpoint between the coils is :

(A) Zero **(B)** $\dfrac{\mu_0 nI}{R}\left(\dfrac{4}{7}\right)^{3/2}$ **(C)** $\dfrac{\mu_0 nI}{2R}$ **(D)** $\dfrac{\mu_0 nI}{R}\left(\dfrac{4}{5}\right)^{3/2}$

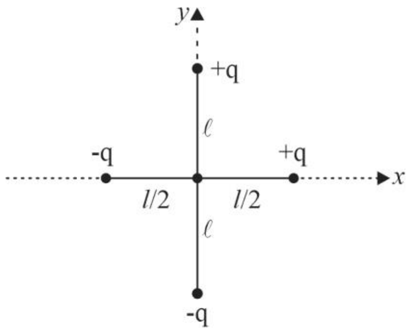
10. For the point midway between the coils (origin):

(A) $\frac{\partial B}{\partial x}=0,\ \frac{\partial^2 B}{\partial x^2}=0$ **(B)** $\frac{\partial B}{\partial x}=0,\ \frac{\partial^2 B}{\partial x^2}>0$

(C) $\frac{\partial B}{\partial x}>0$ **(D)** $\frac{\partial B}{\partial x}<0$

Paragraph for Question 11 - 12

Two insulating rods of length l and $2l$ and mass m each are rigidly fixed at centre. This + structure is placed in x – y plane with point of intersection at origin and rods aligned along x and y axis charge q and -q are attached at the ends of both the rods as shown.



11. Dipole moment of the system is :

(A) $2ql$ **(B)** ql **(C)** $\sqrt{5}ql$ **(D)** $\sqrt{7}ql$

12. If a uniform electric field E_0 is switched on in $+x$ direction. Maximum angular velocity of the structure in subsequent motion is :

(A) $\left(\frac{12qE_0\left(\sqrt{5}-1\right)}{5ml}\right)^{1/2}$ **(B)** $\left(\frac{24qE_0\left(\sqrt{5}-1\right)}{5ml}\right)^{1/2}$

(C) $\left(\frac{24qE_0\left(\sqrt{7}-1\right)}{5\sqrt{7}ml}\right)^{1/2}$ **(D)** $\left(\frac{6qE_0\left(2\sqrt{5}-1\right)}{5ml}\right)^{1/2}$

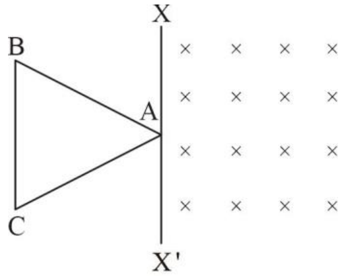
SECTION-2

This section consists of 6 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

1. An isolated conductor initially free of charge is brought in contact with a conducting plate having charge Q. When they are separated the charge on the conductor is $0.6Q$. Charge on the plate is restored to Q after each contact with conductor. Maximum charge that can be given to the conductor this way is found to be αQ. Value of α is ________ .

2. Two charged non-conducting plane discs A and B of radius R and 2R with uniform charge density 2σ & σ $\left(\sigma=6\times10^{-2}\ \frac{\text{coulomb}}{m^2}\right)$ repel each other with a force of 24 newton, when placed at a distance $l=5m$ from each other with their planes parallel to each other. What is the electric flux $\left(\text{in}\ \frac{N-m^2}{C}\right)$ due to plate A through plate B?

3. A triangular loop with each side having length a and resistance R is bordering on a region of uniform magnetic field $B_0 = 5T$. The side BC is parallel to XX'.



It is now pulled into the region of uniform magnetic field with constant acceleration a_0 perpendicular to side BC. Charge flown through the loop (in coulomb) in time $t = \left(\frac{\sqrt{3}a}{2a_0}\right)^{1/2}$ is ________ .

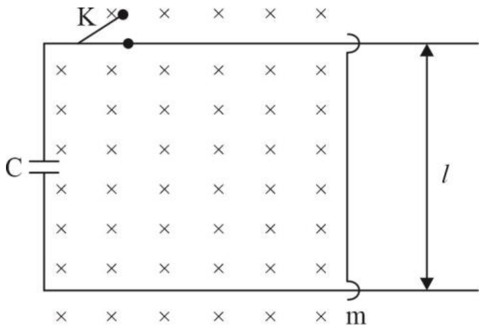
$\left[a = 4m, R = \frac{5}{\sqrt{3}}\Omega, a_0 = 2\frac{m}{s^2}\right]$

4. A thin non-conducting annular disc of inner radius $R_1 = 2m$ and outer radius $R_2 = 5m$ is spinning about on axis through the centre of disc and perpendicular to its plane. Surface charge density on the disc is $\sigma(\sigma\mu_0 = 4$ S.I. units). What is magnetic field at the centre of the disc (in Tesla) if disc is rotating at constant angular velocity $\omega = 5\,\text{rad/s}$?

5. A slider rod of mass m is free to slide on two smooth conducting horizontal parallel fixed rails as shown. Initially key is open and capacitor with capacitance C is charged to a potential V. Key is now closed. What is the maximum speed (in mm/s) of slider rod?

$C = 20\mu F,\ \ B = 100T$

$l = 1m,\ V = 30\,\text{volt}$



6. A current carrying ring has moment of inertia I = 17 kg-m^2 about on axis passing through its centre and perpendicular to the plane of ring. The centre of the ring is at origin. Magnetic moment of the ring is $(3\hat{i} + 5\hat{j})\ A - m^2$. A magnetic field $\vec{B} = (10\hat{i} - 6\hat{j})T$ is switched on at $t = 0$. Maximum angular velocity of the ring (rad/s) in subsequent motion is _______ .

SUBJECT II : CHEMISTRY 60 MARKS

SECTION-1 | Type A

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

1. Which of the following statements are false:

(A) n-type semiconductors are negatively charged

(B) One tetrahedral void per atom is present in HCP structure

(C) In the fluorite structure the cations are located at the lattice points and anions fill all the tetrahedral holes in the CCP crystal.

(D) In the antifluorite structure the cations are located at the lattice points and anions fill the tetrahedral holes in the CCP structure.

2. Two different solutions which are isotonic at room temperature must have the same______. (Assume non-ionizing solute)

(A) Osmotic pressure **(B)** Molarity

(C) Elevation in boiling point **(D)** Depression in freezing point

3. Which of the following statements are correct for SO_2 gas?

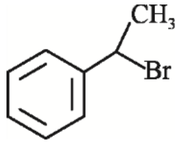
(A) It is used as bleaching agent, disinfectant and preservative

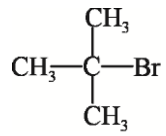
(B) It's molecular geometry is not linear

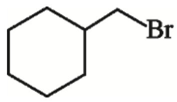
(C) It is colorless, odorless gas and is highly soluble in water

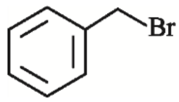
(D) It can be prepared by the reaction of dil. H_2SO_4 with metal sulphide

4. Which of the following statement is correct with respect to nucleophilic substitution reaction?

I. 

II. 

III. 

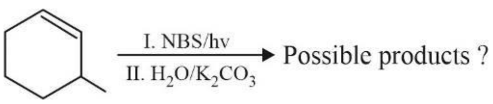
IV. 

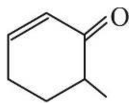
(A) Compounds I and II follow S_N1 mechanism under appropriate conditions

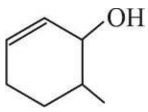
(B) Compounds III and IV follow S_N1 mechanism under appropriate conditions

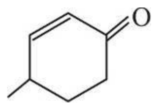
(C) Compounds II and III follow S_N2 mechanism under appropriate conditions

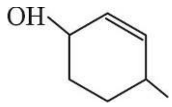
(D) Inversion of optical rotation will occur in compound III as it follows S_N2 mechanism

5. 

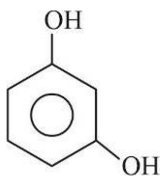
(A) 

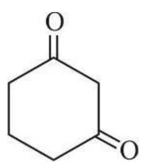
(B) 

(C) 

(D) 

6. Which of the following forms yellow precipitate in iodoform test

(A) 

(B) t-butyl alcohol **(C)** 

(D) Cl_3C-CHO

SECTION-1 | Type B

This section consists of THREE (03) paragraphs. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

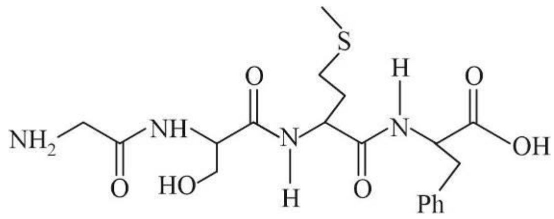
Paragraph for Question 7-8

Amino acids can be classified as $\alpha-, \beta-, \gamma-, \delta-$ and so on depending upon the relative position of amino group with respect to carbonyl group. Amino acids can also be classified as acidic, basic and neutral. Another type of classification can be essential and non-essential.

7. Which is the most basic amino acid.

(A) Lysine **(B)** Arginine **(C)** Histidine **(D)** Leucine

8. The total number of distinct naturally occurring essential amino acids obtained by complete acidic hydrolysis of the polypeptide shown below is



(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

Paragraph for Question 9-10

Heating lead (II) nitrate gives gases 'A' and 'X'. The gas 'X' on cooling changes to colourless solid 'Y'. Solid 'Y' on heating with NO changes to a coloured solid 'Z'.

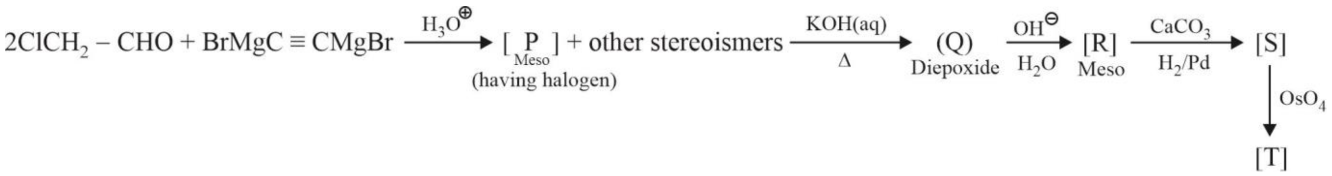
9. Gas 'X' is :-

(A) NO **(B)** N_2O **(C)** N_2O_4 **(D)** NO_2

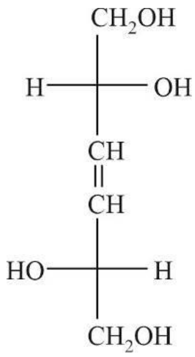
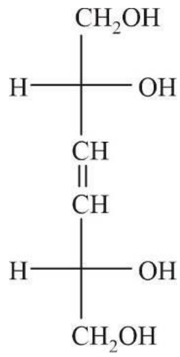
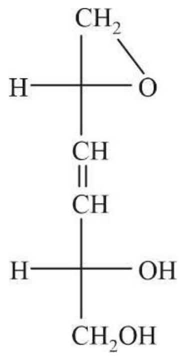
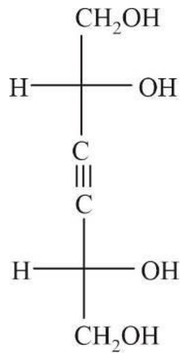
10. Colour of 'Z' is:-

(A) Red (B) Green (C) Blue (D) Yellow

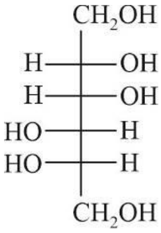
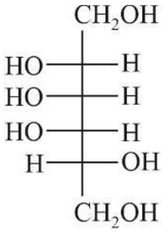
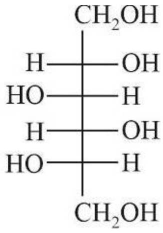
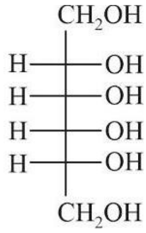
Paragraph for Question 11-12



11. Compound S is ?

(A)  **(B)**  **(C)**  **(D)** 

12. Compound T is?

(A)  **(B)**  **(C)**  **(D)** 

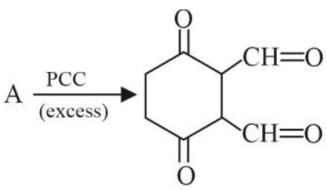
SECTION-2

This section consists of 6 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

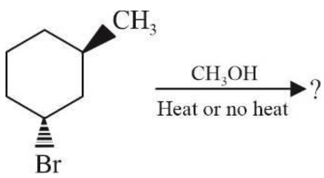
1. In the preparation of HNO_3, (by Ostwald process) we first get NO gas by catalytic oxidation of ammonia. This NO gas further undergoes multiple steps to form HNO_3. Number of moles of HNO_3 produced starting with 12 moles of NH_3 will be:
(Assume 100% yield of each step and ignore the recycling procedure)

2. In F.C.C system, body diagonal length is x and the nearest distance between octahedral void and a tetrahedral void is y. Find the value of x/y.

3. The Van't Hoff factors for KCl, $Al_2(SO_4)_3$ and $K_4[Fe(CN)_6]$ are x, y and z respectively. The value of (y + z)/x is:

4.  (Molar mass = x)

'A' reacts with excess of acetic anhydride and forms 'B' (Molar mass = y). Find the value of $\frac{(y-x)}{22}$.

5. 

What is the sum of possible no. of elimination and substitution products?

6. Sum of chiral carbons present in linear and both cyclic structures of D-glucose is z. The value of (z – 5) is:

SUBJECT III : MATHEMATICS 60 MARKS

SECTION-1 | Type A

This section consists of 6 Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

1. Let $f\left(x^2+y\right)=\left(f(x)\right)^2+f(y)$ for $x, y \in R$, then :

(A) $f(x)$ is odd

(B) $f(x)$ is even $\Rightarrow$ $f(x)=0 \forall x \in R$

(C) $f(x)$ is continuous at $x=0$ $\Rightarrow$ it is continuous everywhere

(D) $f(x)$ is differentiable at $x=0 \Rightarrow f(x)=x\, f'(0), \forall x \in R$

2. If the equation $\left(2^{\frac{1}{\cos^{-1} x}}\right)^{2\pi}-\left(a+\frac{1}{2}\right)\left(2^{\frac{1}{\cos^{-1} x}}\right)^{\pi}-a^2=0$ has only one real solution then subsets of values of 'a' are:

(A) $(-3, 1)$ **(B)** $(-\infty, -3]$ **(C)** $[1, \infty)$ **(D)** $[-3, \infty)$

3. Let $f:\left[0, \frac{\pi}{2}\right] \rightarrow [0,1]$ be a differentiable function such that $f(0)=0,\ f\left(\frac{\pi}{2}\right)=1$, then :

(A) $f'(\alpha)=\sqrt{1-\left(f(\alpha)\right)^2}$ for all $\alpha \in \left(0, \frac{\pi}{2}\right)$

(B) $f'(\alpha)=\frac{2}{\pi}$ for all $\alpha \in \left(0, \frac{\pi}{2}\right)$

(C) $f(\alpha) f'(\alpha)=\frac{1}{\pi}$ for at least one $\alpha \in \left(0, \frac{\pi}{2}\right)$

(D) $f'(\alpha)=\frac{8\alpha}{\pi^2}$ for at least one $\alpha \in \left(0, \frac{\pi}{2}\right)$

4. Let $f(x)=\cos^{-1}(\cos 2x)$ and $g(x)=|\cos x|$ then:

(A) number of solution of $f(x)=g(x)$ in $[0, 2\pi]$ is 4

(B) $\max\{f(x), g(x)\}$ is a periodic function

(C) $\max\{f(x), g(x)\}$ is a non differentiable function for some x,

(D) $\min\{f(x), g(x)\}$ is an even function

5. Consider $f(x)=2\sin x+\sin 2x,\ x \in \left[0, \frac{3\pi}{2}\right]$:

(A) Greatest values of $f(x)$ is 3 **(B)** Greatest value of $f(x)$ is $\frac{3\sqrt{3}}{2}$

(C) Least value of $f(x)$ is zero **(D)** Least value of $f(x)$ is -2

6. The function $f(x)=2|x|+|x+1|-\left||x-3|-3|x|\right|$ has a local minimum or a local maximum at x =

(A) 0 **(B)** $-\frac{3}{2}$ **(C)** $\frac{3}{4}$ **(D)** 3

SECTION-1 | Type B

This section consists of THREE (03) paragraphs. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for question 7 - 8

There exists a matrix Q such that $PQP^T = N$, where $P = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Given N is a diagonal matrix of form

N = diag. (n_1, n_2, n_3) where n_1, n_2, n_3 are three values of n satisfying the equations det. $(P - nI) = 0, n_1 < n_2 < n_3$.
[Note: I is an identity matrix of order 3 × 3]

7. The trace of matrix Q^2 is equal to :

(A) 3 **(B)** $\frac{82}{9}$ **(C)** $\frac{83}{9}$ **(D)** 9

8. The trace of matrix P^{2012} is equal to :

(A) $3^{2011} + 2$ **(B)** 3^{2012} **(C)** $3^{2012} + 2$ **(D)** 3^{2011}

Paragraph for question 9 - 10

Consider $f(x) = \tan^{-1}\left[\frac{(\sqrt{12} - 2)x^2}{x^4 + 2x^2 + 3}\right]$ and m and M are respectively minimum and maximum values of $f(x)$ and $x = a(a > 0)$ is the point in the domain of $f(x)$, where $f(x)$ attains its maximum values.

9. If $\cos^{-1} x + \cos^{-1} y = 3\left(\tan^{-1}\left(\tan\frac{7M}{2}\right) + \tan^{-1}\left(m + \tan\frac{3\pi}{8}\right)\right)$, then $(x + y)$ is equal to :

(A) 2 **(B)** –2 **(C)** 0 **(D)** 3/2

10. If α and β are roots of the equation $x^2 - \left(\tan\left(3\sin^{-1}(\sin M)\right)\right)x + a^4 = 0$, then $\alpha\beta - (\alpha + \beta)$ equals to :

(A) 1 **(B)** –2 **(C)** 3 **(D)** 2

Paragraph for question 11 - 12

Consider a polynomial $y = P(x)$ of the least degree passing through $A(-1, 1)$ and whose graph has two points of inflection $B(1, 1)$ and C with abscissa 0 at which the curve is inclined to the positive direction of x-axis at an angle of $\sec^{-1}(\sqrt{2})$.

11. The value of P(0) is :

(A) $-\frac{1}{2}$ **(B)** $\frac{1}{2}$ **(C)** 0 **(D)** 2

12. The equation $P'(x) = 0$ has :

(A) three distinct real roots
(B) one real root
(C) three real roots such that one root is repeated
(D) none of these

SECTION-2

This section consists of 6 Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

1. Find the sum of an infinite geometric series whose first term is $\operatorname{Lim}_{x\to 0}\sum_{k=1}^{2013}\frac{\left\{\frac{x}{\tan x}+2k\right\}}{2013}$ and whose common ratio is the value of $\operatorname{Lim}_{x\to 0}\frac{e^{\tan^3 x}-e^{x^3}}{2\ln(1+x^3\sin^2 x)}$.

2. Let $f: R\to[-8, 8]$ be an onto function which is given by $f(x)=\frac{bx}{(a-3)x^3+x^2+4}$, $a, b\in R^+$.

If the set of values of m for which the equation $f(x)=mx$ has 3 distinct real solutions is (p, q), then find the value of $(a+b+p+q)$?

3. Let $A=[a_{ij}]_{3\times 3}$, $B=[b_{ij}]_{3\times 3}$ two matrices

where $a_{ij}=\begin{cases}2i+j & 1\le i<j\le 3\\ i^2-j^2 & i=j\\ -(i+2j) & 1\le j<i\le 3\end{cases}$ and $b_{ij}=a_{ji}$ $\forall\ i$ and j,

then find the value of $\det.(A-B)+\det\left((A-B)^2\right)+\det\left((A-B)^3\right)+......\det\left((A-B)^{100}\right)$

4. If α, β and γ are three distinct real values such that $\frac{\sin\alpha+\sin\beta+\sin\gamma}{\sin(\alpha+\beta+\gamma)}=\frac{\cos\alpha+\cos\beta+\cos\gamma}{\cos(\alpha+\beta+\gamma)}=2$ and $\cos(\alpha+\beta)+\cos(\beta+\gamma)+\cos(\gamma+\alpha)=a$, then find the value of $\operatorname{Lim}_{x\to a}\frac{\sqrt{x^2-a^2}}{\sqrt{x-a}+(\sqrt{x}-\sqrt{a})}$.

5. Consider a function $g(x)=3f\left(\frac{x^2+2}{3}\right)+f(6-x^2)\ \forall\ x\in R$ where $f''(x)>0\ \forall\ x\in R$. If $g(x)$ is increasing on $(a, b)\cup(c, \infty)$, then find the value of $a+b+c$.

6. Let $A=[a_{ij}]$ be a 3×3 matrix whose determinant value is 1. A matrix $B=[b_{ij}]$ is formed such that $b_{ij}=\sum_{r=1}^{3}a_{ir}$ $r\ne j$ and $n=\det(A^{-1}B)$ also $A\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}6\\4\\8\end{bmatrix}$ has solutions $x=3, y=2, z=4$ then $2(p+q+r)+n$ where p, q, r satisfy $B\begin{bmatrix}p\\q\\r\end{bmatrix}=\begin{bmatrix}6\\4\\8\end{bmatrix}$.

End of JEE Advanced-5 | PAPER-2 | JEE-2022